

Introduction to Block Codes, Type of Block Codes

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Block Code

Outline

- ❑ Basics of Channel and Codeword
- ❑ Basic Structure of Communication Systems
- ❑ Basics of Block Code
- ❑ Important Parameters of Block Code
- ❑ (4, 3) Block Code {Parity Check}

Basics of Channel and Codeword

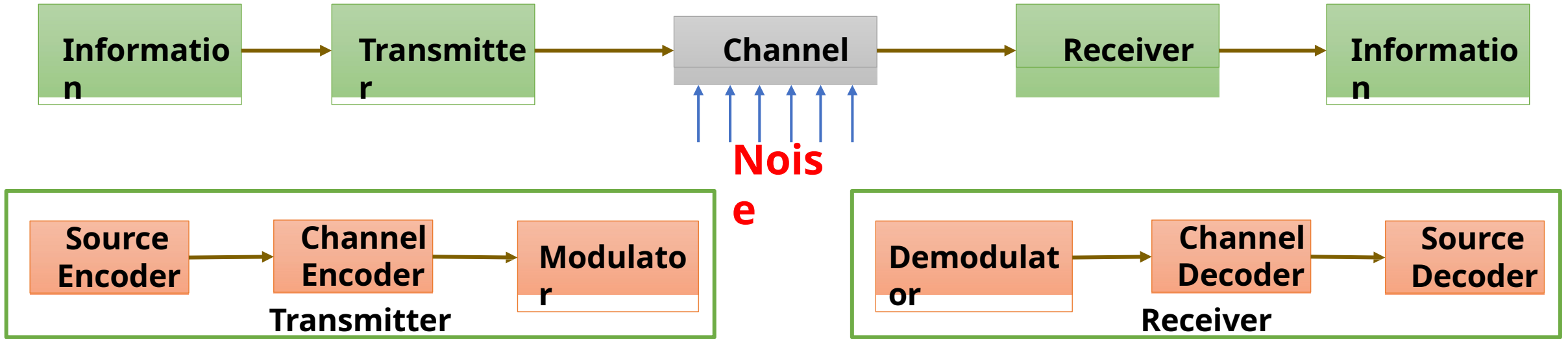
❖ Channel

- In communication systems, the channel is the medium through which the signal travels from the transmitter to the receiver.
 - Examples: air (for wireless), cables, optical fiber, etc.
 - Channels may introduce noise, interference, and distortion, causing errors in transmission.

❖ Codeword

- In digital communication, Information is transmitted as Codewords — specific binary sequences that represent data after encoding.
 - Example: If message bits are $m = 101$, the encoder may convert it to a codeword $c = 101110$.
 - Each codeword follows certain rules designed to detect or correct errors caused by the channel.

Basic Structure of Communication Systems



❖ Source Encoding

- ❑ To remove redundancy from the original message so it can be represented using fewer bits — i.e., to compress the data efficiently.

- Text compression: Huffman coding, Arithmetic coding
- Image/video compression: JPEG, MPEG

❖ Channel Encoding

- ❑ To add redundancy to the transmitted data to make it robust to noise and errors in the communication channel.
- Examples: Linear Block Code (Hamming Code), Convolution Codes, Turbo Codes etc.

❖ **Source Encoding makes data smaller, and Channel Encoding makes data**

Basics of Block Code

- ❑ A Block Code is represented by (n, k) Block Code.
 - n = Number of Bits in codeword
 - k = Number of bits in Information
 - $r = n - k$ = Parity bits or Redundant bits
- ❑ Information $I = [i_1, i_2, i_3, \dots, i_k]$
- ❑ Parity $P = [P_1, P_2, P_3, \dots, P_r]$
- ❑ Codeword $C = [i_1, i_2, i_3, \dots, i_k, P_1, P_2, P_3, \dots, P_r]$
- ❖ **Systematic codes**
- ❑ Systematic codes place message bits at the beginning of the codeword, followed by check bits, making the original message easily identifiable.
- ❖ **Non-Systematic codes**
- ❑ Non-systematic codes intersperse message and check bits, so the original message bits are not located at a fixed position.

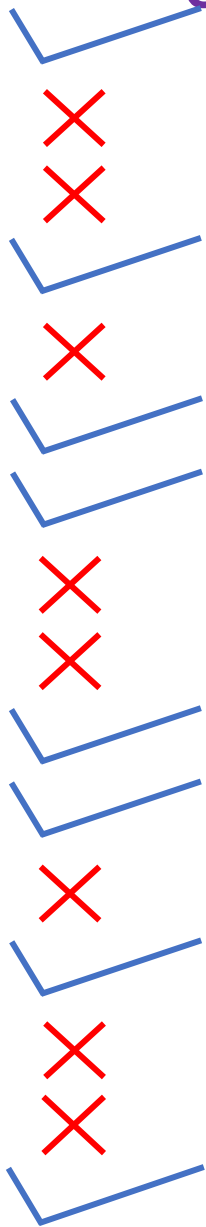
Parameters of Block Code (n, k)

- ❑ Total Codewords in Block code = 2^n
- ❑ Total Codewords for Information in Block code = 2^k
- ❑ Total Redundant Codewords = $2^n - 2^k$
- ❑ Code Rate $R = k/n$
- ❑ Parity bits $P = n - k$

Block Code (4,

Information bits (k = 3 bits)			Parity bit
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

☐ Even Parity Check



Block Code for Parity

Outlines

- ❑ Basics of Block Code for Parity Check
- ❑ Example of Block Code for Parity Check for Encoding
- ❑ Decoding of Block Code for Parity Check
- ❑ Example of Block Code for Parity Check for Decoding

Basics of Block Code for Parity Check

- ❑ Block Code for Parity Check is the simplest form of Error Control used for Error Detection.
- ❑ Block Code for Parity Check can not Correct Error, it can only detect one-bit Error.
- ❑ A parity bit is an additional bit added to make the number of 1's in the codeword even (even parity) or odd (odd parity).
 - Even parity: Total number of 1's is even.
 - Odd parity: Total number of 1's is odd.
- ❑ In (n, k) Block Code for Parity Check, $n = k+1$, means we add one bit of parity (Even or Odd).
- ❑ Information $I = [i_1, i_2, i_3, \dots, i_k]$
- ❑ For Even Parity bit P , $P = i_1 + i_2 + i_3 + \dots + i_k$ {This sum is Mod 2 sum or XOR}
- ❑ For Odd Parity bit P , $P = i_1 + i_2 + i_3 + \dots + i_k + 1$ {This sum is Mod 2 sum or XOR}
- ❑ Codeword $C = [i_1, i_2, i_3, \dots, i_k, P]$

Example of Block Code for Parity Check with Encoding

- **Example 1: Given the (5, 4) Even Parity Block Code, find codewords corresponding to the Information $I = [1\ 0\ 1\ 1]$ and $[1\ 0\ 1\ 0]$**
 - **$(5, 4) = (n, k)$**
 - **Information bits $k = 4$**
 - **Parity Bits $r = n - k = 5 - 4 = 1$**
 - **For Even Parity $P = i_1 + i_2 + i_3 + i_4$ {Sum means XOR Operation}**
 - **Case I: $P = 1 + 0 + 1 + 1 = 1$**
 - **So, Codeword $C = [i_1, i_2, i_3, i_4, P] = [1, 0, 1, 1, 1]$**
 - **Case II: $P = 1 + 0 + 1 + 0 = 0$**
 - **So, Codeword $C = [i_1, i_2, i_3, i_4, P] = [1, 0, 1, 0, 0]$**

Decoding of Block Code for Parity Check

- ❑ At the Receiver stage, we receive n bits codeword V along with the Error.
- ❑ $V = [V_1, V_2, V_3, \dots, V_n]$
- ❑ To determine whether Received Codeword V is a correct codeword, we check the Mod 2 Sum of V.
- ❑ $V = V_1 + V_2 + V_3 + \dots + V_n$ {This sum is Mod 2 sum or XOR}
- ❑ For Even Parity Check, V should be 0; otherwise, there is an error in the received codeword.
- ❑ For Odd Parity Check, V should be 1; otherwise, there is an error in the received codeword.

Example of Block Code for Parity Check with Decoding

- Example 1: Given the (8, 7) Even Parity Block Code, determine whether $V_1 = [10110110]$ and $V_2 = [01101001]$ give parity failures.
 - $(8, 7) = (n, k)$
 - Information bits $k = 7$
 - Parity Bits $r = n - k = 8 - 7 = 1$
 - $P = V_1 + V_2 + V_3 + V_4 + V_5 + V_6 + V_7 + V_8$ {Sum means XOR Operation}
 - Case I: $P = 1 + 0 + 1 + 1 + 0 + 1 + 1 + 0 = 1$
 - For Even Parity $P = 0$, but here we have Parity Failure, so the received codeword is incorrect.
 - Case II: $P = 0 + 1 + 1 + 0 + 1 + 0 + 0 + 1 = 0$
 - For Even Parity $P = 0$, so the received codeword is correct.

Block Code for Product

Basics of Block Code for Product Code

- ❑ A block code with a parity check cannot correct errors; it is only used to detect errors.
- ❑ A block code using a product code can both detect and correct errors.
- ❑ In block code for Product code, two parity-check matrices are combined to form a two-dimensional code structure that allows for both error detection and correction.

Example of Block Code for Product Code

- Example 1: Given the Information I = [1 1 0 1 0 1 1 0 1 0 0 0 0 0 0 0 1 1 1 0 1 0 0 1], Structure the Block Code for Product Code (35, 24).

	Data Bits				Row Parity	
	1	1	0	1	1	
	0	1	1	0	0	
	1	0	0	0	1	
	0	0	0	0	0	
	1	1	1	0	1	
	1	0	0	1	0	
Column Parity	0	1	0	0	1	Overall Parity

- Total 7 rows and 5 columns, so $n = 35$
- Information bits, $k = 24$ bits
- So, given block code is (35,24)
- It is happening as per $(5,4) \times (7,6) = (35,24)$
- It is used to detect and correct a one-

Block Code for Repetition

Basics of Block Code for Repetition Code

- ❑ These are codes that repeat the information bits two or more times.
- ❑ They are a type of block code denoted by $(n, 1)$, where each message bit is encoded into an n -bit codeword.
 - Example: $(4, 1) = (n, k)$
 - $n=4$ and $k=1$
 - Redundancy $r = n - k = 4 - 1 = 3$
- ❑ Here, redundancy is not based on even or odd parity.

Example of Block Code for Repetition Code

❑ Example 1: For (3, 1) Block Code, Explain the Encoding and

Decoding Process.

- $n = 3$ and $k=1$
- Redundancy $r = n - k = 3 - 1 = 2$

❖ Encoding

Data	Redundancy		Codeword		
0	0	0	0	0	0
1	1	1	1	1	1

❖ Decoding Process

- Decoding is done based on

Received Data			Error	Output Data			Info
0	0	0	No	0	0	0	0
0	0	1	One-bit	0	0	0	0
0	1	0	One-bit	0	0	0	0
0	1	1	One-bit	1	1	1	1
1	0	0	One-bit	0	0	0	0
1	0	1	One-bit	1	1	1	1
1	1	0	One-bit	1	1	1	1
1	1	1	No	1	1	1	1

- Majority Vote for (V_1, V_2, V_3) is taken as per Info $I = V_1V_2 + V_1V_3 + V_2V_3$